

Type and Tale: The Effects of Typography and Narratives on Comprehension

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Educational Relevance & Implications Statement

Elaborative comprehension of written text remains essential for learning, even as multimedia options such as video are easily available. Long, complex reading materials are demanding, and learners' attention often fades over time. Our study shows that pairing appealing typography with a narrative frame sustains attention and improves elaborative comprehension. Teachers can support learning from lengthy texts by using professional, or otherwise appealing, layouts combined with embedding content in a narrative.

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Despite the widespread availability of digital learning materials (UNESCO, 2023), students still often opt for printed materials when it comes to reading a text (Raja Jacob et al., 2023; Spencer, 2006). There are good reasons for the continued use of texts because teachers find it more efficient, as the effective implementation of digital materials requires substantial time and cognitive resources (Rice & Ortiz, 2021). However, learning with texts does not go without challenges. When teachers are instructing complex content, they often need to use lengthy texts that result in long learning times for learners to reach comprehension. Many find it challenging to engage with such long texts and lack the motivation to start learning with them (Hidi & Anderson, 1986). Moreover, lengthy texts require learners to keep their attention throughout the learning process (Inzlicht & Schmeichel, 2012). Given these challenges, teachers need strategies to support their students. One promising way is to support learners develop situational interest when learning from lengthy texts (Renninger & Hidi, 2022). Research has shown that heightened situational interest can lead to both stronger overall motivation and sustained learning benefits (Wong & Adesope, 2021).

Building on this argument, teachers using written texts for instruction face the challenge of supporting learners in developing situational interest. Research in emotional design has identified two key components that help in this endeavor: the material must be both visually appealing to trigger situational interest and convey the content's value to maintain situational interest (Endres, Eitel et al., 2024). In written texts, visual appeal is predominantly influenced by typographical features (Spiekermann, 2022). Therefore, a text containing appealing typography should trigger students' situational interest in the material. One way to maintain situational interest is the implementation of narratives. Such narrative tales have been demonstrated to convey value and support maintained situational interest and thus sustained learning (Endres, Eitel et al., 2024; Schneider et al., 2023).

This study analyzes whether visual appeal and narrative tales can motivate learners to start learning and maintain their engagement while learning from lengthy text. More specifically, we

investigated how learning from typographic (i.e. appealing) and neutral texts that are either framed by a narrative or expository examples influence learners' sustained learning outcomes. This study aims to identify key elements to support learning from text in practice, as well as to contribute to the theoretical discussion on the mechanisms of text design on learners' interest development.

Learning from Text

Learning from text is not a passive practice, as it requires active construction processes to reach comprehension. To achieve genuine comprehension, learners create a mental representation in the form of situational models (Kintsch & van Dijk, 1978). Such mental models contain information from the text as well as the prior knowledge the readers possess and apply to interpret the text (Kintsch 1986, 1998). Hence, when learners have understood a text, their mental models consist of both a literal representation at text-level (the text-base) as well as a representation at inference-level corresponding to the situation described in the text (the situational model). In other words, text comprehension is achieved when learners actively construct a coherent mental representation that integrates elements from the text-base with those derived from the generation of inferences (Kintsch, 1998).

Two types of inferences are distinguished (Oakhill et al., 2014). When learners link together elements presented in the text, they achieve comprehension at the *bridging level*. Such bridging inferences are further distinguished into local bridging inferences –for elements close together in the text– and global bridging inferences for elements spread across the text. When learners link together elements from the text with their content-relevant prior knowledge, they achieve comprehension at the *elaborative level*. A successful learning process requires learners to actively make these types of inferences to comprehend the text (Kintsch, 1998). Both bridging and elaborative inferences seem to be the primary source of the effort required to learn from text.

One line of evidence showing that inferences are effortful comes from research comparing learning from digital and printed text. Studies have often demonstrated reduced text comprehension when learners invest less effort, such as when reading on a screen rather than on paper (Delgado et

al., 2018; Salmerón et al., 2023). The shallowing hypothesis (Annisette & Lafreniere, 2017) attributes this difference to reduced mental effort while screen reading compared to print reading. Learners tend to avoid complex strategies when engaging with digital text, which in turn diminishes comprehension (Delgado & Salmerón, 2021).

Another line of evidence as to why text comprehension is an effortful process is that learners who struggle with recognizing words then usually lack the cognitive capacity to invest the effort needed to generate bridging and local inferences. Inference generation during reading is related to reading proficiency. The process demands more effort from young readers (Tricot et al., 2020) but is usually mastered by adolescence (Bianco et al., 2014). In other words, once basic reading abilities have been mastered, it is mostly the generation of bridging (Désiron et al. 2021a, 2021b) and elaborative inferences that predicts successful learning from text.

Following these two types of evidence, the strategies applied to achieve inferences are crucial for learning from text. However, learners often do not apply these strategies spontaneously, especially when their benefits demand a substantial investment of effort. Allocating less effort while reading results in reduced comprehension (Delgado & Salmerón, 2021). It is therefore essential to ensure that learners stay sufficiently motivated to sustain their effort when engaging with lengthy texts when comprehension is the primary educational goal.

Design Elements that Foster Engagement

When instructors use text for learning, they ideally try to motivate their students to engage in high-level inferences. Such motivation is especially relevant when the content is complex, the text is long, and substantial reading time is required (Inzlicht & Schmeichel, 2012). There are many approaches to foster students' motivation. One is to select or design texts motivating learners to invest effort in constructing mental models while reading. For such a design-based approach, emotional design can serve to foster learners' engagement (for meta-analyses see Brom et al., 2018; Wong & Adesope, 2021). Emotional design comprises various elements proven to evoke positive feelings (Brom et al., 2018; Wong & Adesope, 2021) and situational interest (e.g., Endres, Vössing

et al., 2025; Krebs et al., 2024; Schubertová et al., 2023). Some emotional design elements include visual elements, such as presenting instructions in warm colors or round shapes (Désiron & Schneider, 2024; Mayer & Estrella, 2014; Münchow et al., 2017; Plass et al., 2014), using anthropomorphisms (Schneider et al., 2018), or incorporating decorative pictures (Magner et al., 2014; Schneider et al., 2016). Similarly, text-based or auditory elements such as positive words (Stark et al., 2018) and background music (de la Mora Velasco et al., 2023; Endres et al., 2020; Pawar et al., 2020) can enhance learning. Content-related elements such as a narrative frame can also be used. Such content-related features have been found to effectively enhance the value of the learning content and boost both situational interest and learning (Endres, Eitel et al., 2024; Golke & Wittwer, 2024; Beege & Schneider, 2023).

There are two primary explanations for how emotional design benefits learning which are not contradictory but rather address the various challenges learners encounter while reading. The first explanation is formulated in the emotional-design principle (Mayer & Fiorella, 2021; Moreno & Mayer, 2007). This principle suggests that emotional design features aim to enhance learners' motivation to make sense of essential material. This increased motivation initiates goal-directed learning behavior. Moreover, heightened motivation helps the learner sustain such goal-directed behavior. Stronger motivation primes deeper learning processes, which involve investing effort in understanding the learning material. This increased engagement together with the deeper learning strategies ultimately result in better learning outcomes (Mayer & Fiorella, 2021; Moreno & Mayer, 2007).

The second explanation combines the situational interest development theory (Hidi & Renninger, 2006) and self-control motivation theories (Inzlicht & Schmeichel, 2012). The interest-development-self-control explanation suggests that emotional design especially facilitates sustained learning in longer and demanding learning situations (Endres, Vössing et al., 2025). According to this explanation, higher interest levels reduce the need for self-control and thus enhance learning. It assumes that the motivation to engage in self-control declines over time in demanding learning

tasks. Such a reduction in self-control, however, is moderated by the interest learners have in the task (Inzlicht & Schmeichel, 2012). When the design of the learning material supports the development of interest (e.g., by emotional design), it helps learners to stay engaged, particularly under high task demands. To develop interest by the learning material's design, its visual features can increase triggered situational interest (Endres, Eitel et al., 2024). To achieve sustaining learning gains, learners need support in maintaining their situational interest. To ensure sustained learning effects, learners need support maintain their situational interest over time. It is only when learners have reached the level of maintained situational interest that are they more likely to remain committed to demanding learning tasks (Hidi & Renninger, 2006). Maintaining situational interest requires that learners find value in the content (Hidi & Renninger, 2006). This perceived value can be supported by content-related features such as narratives (Endres, Eitel et al., 2024; Golke & Wittwer, 2024). The need for self-control is reduced when visual features foster the development of triggered situational interest and value-related features foster the development of maintained situational interest. As a result, the mental load while learning decreases (Endres, Eitel et al., 2024).

Both explanations assume that visual design features help learners become more motivated to work with text. However, the second explanation adds that additional value might be necessary to attain the crucial level of maintained situational interest to support sustained learning. An additional difference concerns when the benefit might evolve. While the emotional-design-principles perspective suggests that the entire learning process will benefit from emotional design, the interest-development-self-control perspective suggests that it is especially important in later phases of learning.

Emotional Design in Text-Only Learning Material

Most of the research on the emotional design principle has been conducted within the multimedia learning framework (Mayer, 2024). As this framework emphasizes the integration of different representations, studies have predominantly focused on combinations of text (audio or written) and pictures (e.g., Mayer & Estrella, 2014; Plass et al., 2014). However, for certain types of

learning content, the inclusion of additional visual representations may not be beneficial (see also Ainsworth, 2006), as it can trigger redundancy or even become distracting. In this study, we therefore focus on motivating design in text without visual representations. Certain emotional design elements are found even in such text-only materials. We first examined the visual aspects that may trigger situational interest, followed by an exploration of value-related text elements of emotional design.

Research in typography shows that there are many ways to change the visual design of text (Spiekermann, 2022). Two levels of the visual design of text are often distinguished: macrotypography and microtypography. Macrotypography refers to the overall layout including materials, format, fonts, color, and composition, while microtypography concerns fine-grained details such as letter, word, and line spacing (Willberg & Forssmann, 1999). In this paper we focus on the macrotypographic features of the layout, font, and color. These three typographic features are assumed to influence both the cognitive and motivational dimensions of learning from text (Nadolski, 1984).

Most research on the cognitive dimension of reading has focused on text legibility and reading speed. In typography, segmenting the text is a layout element implemented by considering meaningful components of a sentence. In the multimedia learning framework, the segmenting principle suggests the presentation of information in meaningful and coherent segments to manage intrinsic load (Mayer & Fiorella, 2021). An early study by Frase and Schwartz (1979) varied phrase segmentation and indentation as a typographical feature in a series of five experiments with complex procedural texts on the bell system. Although there were no differences in learning outcomes (measured by response accuracy) in either of their experiments, reading time of the text with meaningful sentences segmentation paired with indentation was significantly less than that of the classical text.

On a coarser legibility level, typeface has been found to influence reading speed especially. Faber et al. (2016) showed that manipulating the typeface in a text affected learning outcomes also.

The authors found that an unoptimized typeface such as Comic Sans was more harder to process and in turn, led to better comprehension than an optimized font family such as Arial. Those effects were attributed to cognitive and metacognitive cues evoked by typographically disfluent designs – that is, typefaces that are intentionally less legible (see Kühl & Eitel, 2016). Importantly, studies that focused on more subtle typeface variations such as the presence or absence of serifs, did not find effect on the cognitive dimension of reading speed (Arditi & Cho, 2005). The present study aims to hold the cognitive dimension of typography constant by varying typographic features only within a range neutral to cognitive factors, as supported by educational research and typographic theory.

Regarding the motivational dimension, studies on typeface and font family provide some evidence on font perception. Medved et al. (2023) found that learners perceived round typefaces (e.g., Verdana, Arial Nova) as more pleasant than angular typefaces (e.g., Chaparral Pro, Times New Roman). The issue of font perception also concerns font families (Serif, Sans-Serif, Modern, Monospace, Script/Funny). Some researchers investigated which text characteristics readers attributed to different font families, and which communication purpose was expected from them (Shaikh et al., 2006). For example, Serif fonts are perceived as “stable, practical, mature, and formal” while the Script/ Funny font scored highest for “youthful, happy, creative, rebellious, feminine, casual and cuddly”. Overall, participants reported using Serif fonts preferably for formal documents (e.g., business documents) and Sans-Serif fonts for more casual documents and communication (e.g., website, email).

Overall, this line of research on the aesthetic component of typography shows that visual changes in font seem to have a similar influence on motivation as video and graphical elements such as color or sketched pictograms. Following the explanation of triggering and maintaining interest, the question arises whether typography can also trigger situational interest. In other words, the question arises as to whether typographic elements (text layout, typeface, font style) can play a role similar to that of the usual visual elements of emotional design.

Content Related Features of Emotional Design

With respect to content-related features, the transfer of emotional design elements from multimedia learning materials to text-based materials seems to be relatively straightforward. In multimedia materials, content-related features often include a narrative framework. In text-based materials, analogous approaches have been implemented. Research on text learning has produced a wide body of studies showing that narratives enhance learning outcomes better than expository texts (see the recent meta-analyses by Clinton et al., 2020; Mar et al., 2021). Narratives in texts often serve as advance organizers (Ausubel, 1968, 2000), providing a contextually rich framework that helps learners interpret new information by creating connections to prior knowledge. These meaningful connections facilitate the assimilation of new concepts and improve comprehension (Clinton et al., 2020). Such meaningful connections can also help learners to find value in the learning material. This function seems to be described similarly in both multimedia and text-based learning research and results in comparable effects: Using narratives partly enhance learning outcomes by raising situational interest in both multimedia (Endres, Eitel et al., 2024) and text-based learning (Golke & Wittwer, 2024).

In conclusion, content-related features such as narratives might have similar effects in multimedia and text-based materials. Narratives in text and multimedia learning share the ability to create a contextually rich framework. In this study, we address the question of whether the corresponding elements in both types of learning materials exert a similar effect on developing interest development and on learning outcomes.

The Present Study

The primary goal of this study was to transfer insights from typographical research and emotional design research on learning from texts. In particular, we aimed to analyze their practical utility of typographic design elements when learning from text and to identify the mechanisms underlying potential effects. We tested two specific hypotheses:

We hypothesized a three-way interaction between text design, content frame factor, and the learning phase (*Sustained-Learning-Interaction Hypothesis*). We expected that the combination of typographical design with a narrative frame would yield the most optimal learning outcomes, particularly in the context of sustained learning on both bridging and elaborative comprehension.

Furthermore, we hypothesized that the learning benefits will be mediated by the development of situational interest. More specifically, we expected that typographical design would act as an initial trigger for situational interest, similar to audio-visual features in multimedia research. We expected that the development from triggered situational interest to maintained situational interest would be moderated by the provision of a narrative frame. Lastly, we expected that maintained situational interest would result in enhanced sustained learning (*Interest-Development Hypothesis*, moderated-mediation analysis).

Methods

Design

We conducted an experiment with a 2 x 2 x 2 mixed design. Participants were randomly assigned to one of four experimental conditions, following a Text Design (neutral vs. typographic) x Content Frame (expository vs. narrative) between-subjects design. We also included Learning Phases (first vs. second text section) as a within-subjects factor.

The Text Design factor involved modifications to the text typography to enhance motivation while maintaining legibility, and without directly improving cognitive processing. In the neutral version, the text design followed the default typographic settings typically used in word processing software (e.g., Microsoft Word) and was prepared for printing on A4 paper. In contrast, the typographic version featured a type typography designed to increase the visual attractiveness of the text and was presented as a paperback booklet.

The Content Frame factor involved modifications to the text by implementing either an expository or narrative frame. The expository text frame included factual examples that were not

interconnected. In contrast, the narrative frame integrated structurally equivalent examples into a storyline featuring a protagonist.

The Learning Phases factor distinguished between content introduced in the learning material's first and second sections. This factor was operationalized by assigning specific learning content to corresponding posttest questions: half of the questions corresponded to content from the first section and the other half to content from the second section. Note that the content introduced in the first section was not required to understand that presented in the second section.

Participants

We used G*Power (Faul et al., 2009) for an a-priori power calculation. As some of our hypotheses focus on the two between-subjects factors only, we conducted an a priori power analysis for 2x2 between-subjects comparisons. To be able to identify a medium effect size ($\eta^2 = 0.06$), as in previous research on emotional design, (Wong & Adesope, 2021) the power calculation resulted in a recommended sample size of 128 (α -level = 0.05, power = 80%).

Initially, 141 participants from two grammar schools completed the study. Nine participants were excluded from our analyses because they withdrew consent or did not fully complete the experiment. Hence, the data from 132 participants (60 females, 71 males, 1 outside binary), with a mean age of 16.04 years ($SD = 1.07$) were analyzed.

The study was approved by the local ethic committee at the university of *anonymous*. We followed the approved procedure in our study.

Instructional Text

We presented a text on sleep in all conditions. The content was divided into two sections to enable analysis of the Learning Phase's factor. The first section was about sleep in general and individual sleep patterns, and the second section was about the functions of sleep.

Operationalization of the Text Design Factor

For the Text Design factor, we varied different typographical elements including the general layout as well as typeface (i.e., font family) and font style (e.g., italics, bold). The text's typography

was modified to enhance motivation without influencing cognitive processing (e.g., legibility). The typography of the neutral design used a design set as default in word processing software such as Microsoft Word. The text was printed as an A4 portrait layout in 10-points Time New Roman Serif font, with 1-point line spacing formatting (see Figure 1). Additionally, in the neutral design, key elements for learning outcome were signaled by applying bold and italics together.

The typographic design was created by a typographer and presented in a bound B-format paperback booklet. For the body of the text, changes included using the Garamond Serif font in 9.5-points and increasing line spacing to 1.5. The Garamond Serif font in 10-points italics was used for second-level headings (see Figure 1). Additionally, the Yanone Kaffeesatz sans Serif font was used in bold for the title (72-points, black) and first-level headings (11-points, brown). Key elements were highlighted by bold only. Text sections pertaining to results from experiments were presented in Yanone Kaffeesatz font (10-points, regular).

Operationalization of the Content Frame Factor

In the expository version of the Content Frame, all explanations referred to emotionally neutral expository examples, and the text was worded factually. In the narrative framework version, examples with equivalent structures were embedded within an overarching narrative. The narrative was split between the two-text sections. In the first section of the text, Lisa, the narrative's protagonist, goes on vacation with her family. In the second section of the text, Lisa begins an internship in a sleep lab. Headings were also modified to correspond to Lisa's question on the topic addressed. Changes in content were controlled for in terms of length (less than a five percent difference) and reading difficulty (Flesch of 50 with an expository frame, 52 with a narrative frame).

***** Place Figure 1 near here *****

Operationalization of the Learning Phase

The Learning Phase factor was operationalized in the text by differentiating between the knowledge that could be gained in the first or second section in the text. The content of the first part

was not needed to understand the second part. The lengths of the first section (expository frame: 2097 words, narrative frame: 2113 words) and second section (expository frame: 2069; narrative frame: 2262) were similar.

Learning Outcomes

Learning outcomes included open-ended questions targeting comprehension as indicated by bridging and elaborative inference questions. We looked separately at the questions about the contents from the text's first and second sections. The questions tapping on the second part of the text operationalize the sustained effect of emotional design (see Endres et al., 2020; Münchow & Bannert, 2019).

Bridging comprehension was assessed by having participants generate local or global bridging inferences, for example by linking information spread out in the text about a specific process (e.g., "Name the first two stages of sleep and describe their key characteristics). Two items assessed bridging comprehension of content from the first section and two items from the second section. Answers were coded on a 0–28 point scale for each section. To ensure objectivity, 20% of the responses were randomly selected and double-coded. Inter-rater reliability for the first section ($ICC = .80$) and second section ($ICC = .83$) was good. Two bridging comprehension scores were computed as percentage correct, one for each section.

Elaborative comprehension was assessed by having participants explain or compare content from the text (e.g., "Define sleep-relevant time cues and give examples. Explain why these are important based on the bunker experiments."). Three items assessed elaborative comprehension of the first section, and three items from the second section. Answers were coded on a 0–28 point scale for each section. To ensure objectivity, 20% of the responses were randomly selected and double-coded. The inter-rater reliability for the first section ($ICC = .90$) and the second section ($ICC = .99$) was good to excellent. Elaborative comprehension scores were computed as percentage correct for each section.

Measures

Objective Prior Knowledge

To assess objective prior knowledge, we used a free-recall question that asked learners to write down everything they knew about the functions and processes of sleep. The same coding scheme as that applied to the learning outcome variables was used. To ensure objectivity, 20% of the responses were randomly selected and double-coded. Inter-rater reliability was high (ICC = .94). Prior knowledge is presented in percentage.

Subjective Prior Knowledge

Subjective prior knowledge was assessed using six topic related self-report items on themes presented in the materials (e.g., healthy sleep). Three items were linked to each text section, ensuring equal number of questions for both sections. Responses were given on a 7-point Likert scale (1 = very poor to 7 = very good; Cronbach's $\alpha = .83$).

Perceived Mental Load

We assessed mental load as a multidimensional construct that can hinder learning (Endres, Bender et al., 2025). To capture the mental load from reading, we adapted three items from Endres et al. (2020). To capture it from the material's format, we adapted three items from Klepsch et al. (2017). The overall scale's internal consistency was good, Cronbach's $\alpha = .86$ ¹.

Existing Interest

We assessed existing interest with five items adapted from Knekta et al. (2020, e.g., I think sleep is very interesting). Internal consistency was good, Cronbach's $\alpha = .88$.

Situational Interest

To measure triggered and maintained situational interest, we adapted the 12 items developed by Endres et al. (2020), in which triggered situational interest was associated with learners' feeling

¹ As we did not formulate specific hypotheses for these variables, we did not analyze them within the scope of this manuscript. Interested readers can access the data on OSF.

toward the design and maintained situational interest was associated with learner's feeling and value toward the content. Participants were to rate the items on a 7-points Likert scale (1 *does not apply at all* to 7 *applies fully*). Scale reliability was good, with triggered situational interest, Cronbach's $\alpha = .73$, and maintained situational interest, Cronbach's $\alpha = .84$.

Monitoring

To assess learners' metacognitive monitoring, we assessed the judgment of learning (JOL). We requested learners to rate their ability to answer questions such as those posed in the pretest. The rating scale for these JOLs ranged from zero to ten¹.

Demographics

To test comparability between groups, we assessed demographic variables. Age, gender, class level, and biology grade were measured using separate single-choice items.

Procedure

The experiment took place during a 90-minute class session. After completing measures for prior knowledge and a scale for existing interest, participants were randomly assigned to read the text on sleep in one of the four experimental conditions: neutral-expository framework ($n = 33$), neutral-narrative framework ($n = 31$), typographic-expository framework ($n = 34$), typographic-narrative framework ($n = 34$). We allocated 25 minutes for reading the text in class. Students were instructed to use the entire period for this task. After 25 minutes, all students proceeded with the study. After learning with the text, the participants completed a series of scales measuring situational interest and mental load. Students answered the single JOL item. After the process measures were assessed, the students worked on the learning outcomes questions.

Results

We set the alpha level at .05. We relied on two-sided tests for all analyses. We relied on η^2 as an effect size index. The values 0.01, 0.06 and 0.14 were considered as small, medium, and large effects size, respectively. We used the process tool (Hayes, 2022) for our moderation and mediation analyses.

Prerequisite Analysis

We conducted Bayesian analysis to gain evidence for anticipated null effects between our randomized groups in demographic data and learning prerequisites. We employed the conservative JZS prior and a Prior scale of .0707 (Morey & Rouder, 2015) to ensure conclusive evidence for null effects. Using such a prior raises the likelihood that the Bayes Factor will yield insufficient evidence rather than falsely confirming the null hypothesis. The values of the Bayes Factor (BF_{01}) 1-3, 3-10, 10-30, 30-100, and over 100 were considered as anecdotal, moderate, strong, very strong and extreme evidence, respectively (Lee & Wagenmakers, 2014).

We conducted a factorial Bayesian ANOVA across all four conditions to assess whether randomization resulted in comparable groups. In the Bayesian ANOVAs, we found evidence supporting the null hypothesis regarding our demographic and prerequisite variables between groups (age: $BF_{01} = 18.24$, biology grade: $BF_{01} = 23.14$, existing interest before the intervention: $BF_{01} = 15.58$, objective prior knowledge: $BF_{01} = 1.40$, subjective prior knowledge: $BF_{01} = 14.55$). As the null hypothesis for prior knowledge is only anecdotal, we will use it as a potential covariate in a robustness test. The employed Bayesian contingency table showed evidence for the null hypothesis for gender ($BF_{01} = 4.27$) and class level ($BF_{01} = 4.86$). Descriptive statistics for learning prerequisites are found in Table 1.

***** Place Table 1 near here *****

We additionally conducted a stepwise regression analysis to predict bridging and elaborative comprehension outcomes, incorporating all available demographic data and learning prerequisites. We found that objective prior knowledge ($\beta = .43, p < .001$) and biology grade ($\beta = -.18, p = .023$) significantly predicted bridging comprehension. For elaborative comprehension, only objective prior knowledge proved to be a significant predictor ($\beta = .44, p < .001$). To enhance the robustness of our results, we will apply both prior knowledge and biology grade as covariates in our robustness analysis.

Sustained-Learning-Interaction Hypothesis

We hypothesized an interaction between Text Design and Content Frame with respect to both learning outcomes (bridging and elaborative comprehension as separate depended variables) over time (interaction with the Learning Phase). The mixed-design MANOVA revealed two significant main effects, Text design: $F(2,127) = 4.56, p = .012, \eta_p^2 = .07$; Content Frame: $F(2,127) = 5.85, p = .004, \eta_p^2 = .08$. However, these main effects were, as hypothesized, qualified by an interaction between Text Design and Content Frame, $F(2,127) = 3.71, p = .027, \eta_p^2 = .05$. This interaction of the between-subjects variables yielded the highest learning outcomes when typographical design and narrative frame were combined.

The Learning Phase did not moderate the interaction between Text Design and Content Frame, $F(2,127) = 2.00, p = .438, \eta_p^2 = .01$. The descriptive statistics for our key variables are presented in Table 2.

***** Place Table 2 near here *****

Examining the two learning outcomes separately, bridging and elaborative comprehension, revealed that elaborative comprehension was the decisive variable for the interaction between Text Design and Content Frame we had previously observed. There was no significant two-way interaction on bridging comprehension, $F(1,128) = 3.10, p = .081, \eta_p^2 = .02$, but on elaborative comprehension, $F(1,128) = 7.35, p = .008, \eta_p^2 = .05$. Descriptively, we identified the highest scores for both learning outcomes variables when typographical design and narrative frame were combined.

Within our MANOVA, we moreover observed a significant two-way interaction between the within-subject factor, Learning Phase, and the Content Frame between-subjects factor, overall $F(2,127) = 3.71, p = .027, \eta_p^2 = .05$. Our participants' learning decline over time was smaller with the Narrative Frame than with the Expository Frame. Learners in the Narrative Frame condition outperformed learners in the Expository Frame condition in the second section of the learning phase. When examining the two measures of learning outcomes separately, we found that the

driving force for this interaction was the bringing comprehension, $F(1,128) = 5.90, p = .017, \eta_p^2 = .04$. This interaction showed that the decrease in the generation of bridging inferences was smaller with the narrative frame than with the expository frame. Although the trend was the same in terms of generating elaborative inferences, it did not reach the level of statistical significance, $F(1,128) = 3.17, p = .077, \eta_p^2 = .02$, see Figure 2.

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Robustness Check

To assess the robustness of our findings, we conducted the same MANOVA by incorporating the objective prior knowledge and biology grade covariates. The pattern of results remained consistent with our main analyses (see Sustained-Learning-Interaction Hypothesis section). Specifically, we found again a significant interaction between Text Design and Content Frame, $F(2,125) = 3.64, p = .021, \eta_p^2 = .06$. However, the three-way interaction was not significant, $F(2,125) = 0.78, p = .459, \eta_p^2 = .01$.

Examining the two learning outcomes bridging and elaborative comprehension: these also yielded a consistent pattern regarding the between-subject interaction, bridging comprehension: $F(1,126) = 3.28, p = .072, \eta_p^2 = .03$; elaborative comprehension: $F(1,126) = 7.66, p = .007, \eta_p^2 = .06$. Similarly, the interaction between the within-factor, Learning Phase, and the Content Frame between-factor remained consistent, bridging comprehension: $F(1,126) = 7.94, p = .006, \eta_p^2 = .06$; elaborative comprehension: $F(1,126) = 3.28, p = .073, \eta_p^2 = .03$.

Interest-Development Hypothesis

To test our Interest-Development Hypothesis, we conducted a moderated mediation analysis. Our assumption was that typographical design would increase triggered situational interest, which would then increase the maintained situational interest. This increase between situational and maintained situational interest would be moderated by the Content Frame Factor. Greater maintained situational interest should result in better elaborative comprehension in section two. Our analysis, however, failed to support this hypothesis. Specifically, the initial path linking

Text Design to triggered situational interest was not statistically significant, $a: b = .31, p = .15$ CI^{95} $[-.11, .74]$. As a result, the indirect effect also lacked significance, $b_{\text{expository Frame}} = 0.80, CI^{95} [-0.68, 2.40]$, $b_{\text{narrative Frame}} = 0.69, CI^{95} [-0.60, 2.27]$. The typographical design we employed did not trigger situational interest.

Given that the interaction between Text Design and Content Frame r proved to increase comprehension, we conducted an additional exploratory sequential mediation analysis for the combined group. This analysis aimed to explore whether the expected underlying processes of interest development were responsible for the combined benefit instead of the moderated benefit as assumed in our hypothesis. We employed both text design and content frame as a combined contrast condition $(-1, -1, -1, 3)$, using triggered and maintained situational interest as sequential mediators and comprehension as the outcome variable. This analysis revealed that such a contrast increased triggered situational interest, subsequently boosting maintained situational interest, and ultimately leading to enhanced sustained elaborative comprehension, indirect effect $\beta = 1.69, CI^{95} [0.27, 3.55]$ (see Figure 3). Our data suggests that the synergistic effect of typographical design and narrative frame was essential to initiate the process of interest development. Maintained situational interest did not predict bridging comprehension, $b = 1.69, p = 0.28$; resulting in an insignificant indirect effect, $\beta = 0.05, CI^{95} [-0.04, 0.016]$.

***** Place Figure 3 near here *****

Discussion

The primary goal of this study was to transfer the insights from typographical research and emotional design research on learning from texts. Our study aimed to analyze the practical utility of typographic design elements in text-only learning material, and to identify the mechanisms underlying their potential effects. To this end, we implemented two between-subjects conditions: varying text design and utilizing either a narrative or expository frame. We assessed inferential comprehension at both the bridging and elaborative levels for both the first and second sections of the learning material.

The Sustained-Learning-Interaction Hypothesis. We expected to detect a three-way interaction: Typographical design combined with a narrative frame improved learning outcomes, especially in the second section of the learning material. Our results indicated that a typographical design combined with a narrative frame did indeed result in the highest learning outcomes; specifically on elaborative comprehension outcomes. However, our data did not support the sustained learning interaction with the text sections that we had anticipated .

This result highlights a key difference between our study on typographical design and earlier findings in the connected emotional design literature. In Emotional Design, more pronounced effects were identified with (at least) medium learning durations (Wong & Adesope, 2021). In this study, we detected an overall learning benefit from text-only material starting already in the first text section. There are several tentative explanations for this main effect on elaborative comprehension. One possibility, consistent with self-control theory, is that reading a complex text as in our study demands a lot of effort from learners and therefore requires substantial self-control already in early learning phases. This earlier onset of the need for self-control when learning from text might trigger an earlier onset of the motivational shift (Inzlicht & Schmeichel, 2012). If this effect is replicated in future studies, the use of emotional design in text may be even more important than in multimedia materials such as videos.

A second tentative explanation is that the difference in the transience of materials in videos and texts plays a decisive role (see, e.g., Wong et al., 2012; Singh et al., 2012). While reading, learners can return to earlier passages, which reduces the immediate processing demands that are present when learning with videos. When learners encounter difficulties with self-control while reading, they may more intuitively stop processing or return to reread missed passages. However, such metacognitive monitoring and regulation impose additional mental demands (Seufert, 2018). These demands may even amplify the effects of interest development. When learners are more interested, they may find learning less demanding (Endres et al., 2020) and feel more motivated to engage in additional regulatory processes (Hidi & Renninger, 2006). The possibility of remedial

learning in texts may explain why there is no interaction between the design and learning phase factors.

Future studies might investigate which of these tentative explanations can attract more support through more fine-grained analyses of learning timing and other process measures. We based our decision (to use two sections, each lasting around 10 minutes) on evidence from research on multimedia design (Wong & Adesope, 2021) and instructional videos (Endres et al. 2020). However, if the crucial processes involved in reading are different from learning from video, we need to investigate these processes in more detail to see if self-control over time also plays a role in learning from text, or whether a different process is involved in this context.

Interest-Development Hypothesis. We found that both typographical design and narrative were necessary to initiate the development of triggered and situational interest (Hidi & Renninger, 2006). However, Text Design and Content Frame failed to demonstrate the selective effects we had expected on triggered and maintained situational interest. Learners did not perceive the changes in visual typographical design as strongly as they did in similar research with multimedia material. Contrary to findings in multimedia research (Endres, Eitel et al., 2024), the visual effect of typographical design alone did not evoke triggered situational interest.

Despite the typographical differences implemented between the two text design versions (neutral and typographical design), we found that triggered situational interest was not notably influenced by typography alone. There are several tentative explanations for this finding. One possibility is that young learners are less accustomed to diverse typographical forms. With the growing digitalization of reading materials, learners can often choose their font and font size, which may limit their exposure to other typography styles. As the literature on digital versus print reading indicates, preferences and beliefs about reading have evolved over recent decades, reflecting changes in reading behavior (Delgado et al., 2018). More research comparing self-selected and preselected typography could shed light on this tentative explanation and identify potential age differences.

Another explanation might be the less salience of typographical changes compared to the implementation of visual emotional design elements in multimedia material. While changes in font and spacing are notable, altering the color of a large graphic in multimedia materials is probably much more noticeable. For example, in explainer video research (e.g., Endres et al., 2020) and text-picture instructional documents (e.g., Désiron & Schneider, 2024), changing the entire background or its color creates differences that might be more salient to learners. It is possible that the typographical changes in this study were too subtle to evoke significant differences. Future studies could investigate factors such as background color to enhance the salience of typographical changes.

Selective Effects on Bridging and Elaborative Comprehension

Analyzing the two levels of comprehension in our study revealed interesting selective effects. The interaction of the main effects of Text Design and Content Frame seemed to be driven primarily by the elaborative level of comprehension. This level of comprehension involves elaborate inferences that connect the written content to the learners' prior topic knowledge. Interestingly, for this higher level of inference, both triggered and maintained interest were necessary to achieve improved learning gains. These findings highlight that a deep understanding requires the motivation provided by maintained situational interest to lead to learning benefits (Renninger & Hidi, 2020).

Further, we identified interaction between the Content Frame factor and Learning Phase on bridging comprehension. When a narrative frame was implemented, bridging comprehension for the second section remained higher than without a narrative frame, indicating a sustained effect of the content frame factor on bridging comprehension. Following the previous argument on why we observed a main effect in this study, it might be that bridging inferences are somewhat easier for learners, and thus still easy enough for all learners reading the first section of the learning material. As the motivational shift for those easier inferences starts later, the interaction over time might be more pronounced. Typographical design alone did not seem necessary to achieve this level of

inferential comprehension. Overall, this pattern of results suggests that a narrative's effectiveness differs when its in a text or in a video, which might be attributable to the reading process. When learners read, they activate various schemata early in the learning process. The use of a narrative frame immediately influences the choice of schemata applied to construct the situation model. As reading in our study is self-regulated, learners were able to take the time they needed to update their schemata when it did not exactly match the used narrative. Therefore, narratives may play a greater role not only in fostering interest development, but also in constructing the mental model.

Overall, the selective effects observed in our study revealed the same direction across different measures of comprehension. Although this differentiation is theoretically interesting, our results indicate a difference in magnitude, not direction. Therefore, practical recommendations regarding selective learning effects can only focus on specific features. We detected no negative effects of using narratives or typographical design on any comprehension level.

Limitations

In our study, we varied multiple features of typography simultaneously due to the interplay of formatting features such as layout, typeface, and font style. We chose this approach to reflect real-world conditions, as high-level typography is based on the assumption that certain features systematically co-occur or mutually influence each other (Spiekermann, 2022). Future research could adopt a less ecological approach by, for example, varying only fonts. However, our findings suggest that a single typographic component may not yield significant differences. As shown, typographic design alone appears less capable of triggering interest, even in the combination we tested. Thus, we anticipate even smaller effects in studies examining these aspects individually.

Another limitation of our study is the exclusion of variables commonly examined in text comprehension research, such as reading abilities or distinctions in inference levels. We did not expect such individual differences to moderate our results. Future studies interested in individual differences might replicate our findings while assessing reading abilities.

Theoretical Implications

In our study, neither of the two theoretical explanations (i.e., the emotional-design principle and interest-development-self-control explanation) aligned perfectly with our data. The emotional design principle posits that positive feelings evoked by visuals enhance overall learning. However, our results show that overall learning benefits only occurred with the interaction between visual and content-related motivational design factors. It seems necessary for emotional design elements to both evoke positive feelings and support the development of value. This need for value aligns with the interest development theory (Hidi & Renninger, 2006) and the interest-development-self-control explanation (Endres et al., 2020).

Conversely, the interest-development-self-control perspective predicted that both factors are required to develop interest, and that their effect would interact with the text section factor. Our data did not however reveal a significant interaction with the text section factor. Although the effect's direction was correct, it was smaller than in learning from instructional videos. We observed the interaction over time exclusively with the narrative on bridging comprehension.

Our data suggest that both mechanisms play a role in learning from text. Stronger general motivation appears to be evoked by combining visual and content-related factors, enhancing elaborative comprehension and reflecting the predictions of both the emotional-design principle and interest development theory. We also observed an interaction over time on bridging comprehension. The different mechanisms seem to influence diverse learning outcomes. Future studies should apply more precise process measures to determine which motivational process predominates for each learning outcome. For instance, assessing the learning process in greater detail, for example by using a finer time scale could provide substantial insights. In our ecologically valid learning scenario, we did not collect online learning measures such as eye-tracking data during the reading process. Future research could employ online measures for a more detailed analysis of different inference measures in controlled studies to identify the conditions under which more elaborated inferences are built.

Practical Implications

Our study shows that combining typographical design with narrative frame improves comprehension. The narrative frame helps learners keep their attention to details in the text. This combination boosts comprehension, especially at the level of elaborative comprehension. Teachers should use whenever feasible professional typeset texts and narratives to encourage students to employ effortful learning strategies necessary for high comprehension levels.

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Table 1*Descriptive statistics for prerequisite sample characteristics.*

	Neutral Design		Typographical Design	
	Expository Frame (<i>n</i> = 33)	Narrative Frame (<i>n</i> = 31)	Expository Frame (<i>n</i> = 34)	Narrative Frame (<i>n</i> = 34)
Gender (in % female)	64%	42%	62%	48%
Classlevel (9th – 12th grade)	7/18/4/4	6/18/5/2	6/19/5/4	3/20/7/4
Age*	16.12 (1.27)	15.94 (0.81)	15.97 (1.00)	16.12 (1.15)
Biology grade*	2.40 (1.22)	2.26 (1.31)	2.40 (1.22)	2.33 (0.85)
Objective prior knowledge*	18.04 (10.42)	21.45 (11.45)	19.77 (8.21)	24.91 (12.54)
Subjective prior knowledge*	3.67 (1.05)	3.61 (0.92)	3.86 (1.02)	3.81 (0.84)
Existing interest*	2.46 (1.21)	2.34 (1.36)	2.38 (1.19)	2.37 (0.91)

Note. *Values are means and standard deviations (in parentheses)

Table 2

Descriptive statistics for dependent variables regarding our hypotheses.

Variable	Neutral Design		Typographical Design	
	Expository Frame (<i>n</i> = 33)	Narrative Frame (<i>n</i> = 31)	Expository Frame (<i>n</i> = 34)	Narrative Frame (<i>n</i> = 34)
<i>Situational Interest</i>				
Triggered Situational Interest	4.19 (1.30)	4.25 (1.21)	4.26 (1.28)	4.81 (1.13)
Maintained Situational Interest	5.25 (1.02)	5.31 (0.93)	5.15 (1.22)	5.64 (0.90)
<i>Bridging comprehension</i>				
Bridging comprehension- First section	36.46 (15.65)	35.84 (16.67)	36.98 (12.30)	42.86 (13.62)
Bridging comprehension - Second section	21.61 (9.25)	27.81 (11.91)	22.78 (10.77)	35.43 (14.45)
<i>Elaborative comprehension</i>				
Elaborative comprehension - First section	27.44 (10.01)	26.65 (9.46)	28.28 (11.49)	32.89 (9.08)
Elaborative comprehension - Second section	22.59 (9.59)	23.13 (9.78)	22.44 (11.38)	34.30 (16.75)

Note. Values are means (percent correct) and standard deviations (in parentheses)

Figure 1

Example pages from the learning material with narrative frame in neutral (left) and typographic (right) Text Design.

oder Lisa, die auf der anderen Seite des Bettes schläft, hoch. Bei der sogenannten REM-Schlaf-Verhaltensstörung ist der Körper nicht gelähmt. Als Lisa das beim Frühstück erwähnt, erzählt ihre Oma, dass Lisa Opa manchmal in der Nacht wild um sich schlagen würde. Deshalb schlafen sie inzwischen in getrennten Betten, damit sich beide nicht verletzen.

In der REM-Schlafphase nimmt unsere Atemfrequenz wieder zu und unser Blutdruck sowie unsere Herzfrequenz steigen an. Zum Ende der Schlafperiode wachen wir entweder aus dem REM-Schlaf auf oder gleiten wieder in einen neuen Schlafzyklus.

Lisa weiß inzwischen, dass unsere REM-Phasen im Laufe der Nacht länger und die Tiefschlafphasen kürzer werden. Zwischen den einzelnen Schlafphasen bemerkt sie, dass Mona immer wieder aufwacht. Lisa weiß, dass das oft nach dem Durchlaufen eines Zyklus passieren kann. In der Regel ist Mona nur kurz wach und schläft direkt wieder ein. Auch auf Lissas Nachfrage kann sich Mona am nächsten Tag nicht mehr an das Wachsein erinnern. Lisa kann sich jetzt erklären, wieso sich der Schlaf ihrer Schwester über die Nacht in seiner Qualität verändert, aber sie fragt sich immer noch, warum sich ihr eigener Schlaf von Monas oder dem von ihren Cousins Luka und Toni unterscheidet, daher schaut sie nach, was den Schlaf beeinflusst.

Was beeinflusst unseren Schlaf?

Lisa findet heraus, dass die Regulierung von Schlaf mit diesen Unterschieden zu tun haben kann. Das dominierende Leitmodell für Schlafregulierung ist heute das **Zwei-Prozess-Modell**. Das Modell enthält den **homöostatischen Prozess "Faktor S"** und den **zirkadianen Prozess "Faktor C"**. Der homöostatische Prozess beeinflusst unsere Schlafintensität, der zirkadiane Prozess unsere Schlafdauer.

Schlafintensität (homöostatischer Prozess - Faktor S) **Schlafdauer** (zirkadianer Prozess - Faktor C)

Die beiden Prozesse arbeiten manchmal zusammen und manchmal gegeneinander. Lisa erfährt, dass sie durch ihr Verhalten die Prozesse beeinflussen kann. Wenn Lisa z. B. Stress, Angst oder einen Langstreckenkflug hinter sich hat, hat das eine Wirkung auf beide Prozesse. Somit beeinflussen auch externe Faktoren den Schlaf.

1. Homöostatischer Prozess

Homöostase beschreibt ein **dynamisches Gleichgewicht**. Ein Beispiel für ein solches Gleichgewicht ist unsere Körpertemperatur, sie wird in einem homöostatischen Prozess geregelt – Abweichungen werden sofort gegen korrigiert.

Bei der Schlaf-Homöostase geht es um den quantitativen Bedarf an Schlaf, der sich im Wachzustand aufbaut und im Schlaf abnimmt. Dieser homöostatische Druck schlafen zu müssen, hängt nicht nur davon ab, wie lange Lisa wach ist, sondern auch von ihrer Aktivität im Wachzustand. Ist Lissas Schlafdruck abends hoch genug, weil sie zum Beispiel den ganzen Tag in den Bergen wandern war, kann sie leicht einschlafen.

Wenn Lisa nach einer durchgefeierten Nacht Schlafmangel aufholt, wird die Zeit, die sie danach zusätzlich im Schlaf verbringt, als Erholungsschlaf bezeichnet. Dies ist eine Folge des homöostatischen Prozesses.

2. Zirkadianer Prozess

Unter dem Begriff „zirkadianer Rhythmus“ ist ein **biologischer Rhythmus mit einer Dauer von etwa 24 Stunden** gemeint (aus dem Lateinischen für ungefähr einen Tag). Ein typischer zirkadianer Rhythmus ist also unser Schlaf-Wach-Rhythmus.

Lisa begreift, dass der zirkadiane Prozess dabei hilft, den homöostatischen Prozess in Einklang zu halten. Er hilft Lisa dabei, die Nacht durchzuschlafen und tagüber wach zu bleiben. Ohne einen zirkadianen Prozess, der 24 Stunden dauert, würde der homöostatische Druck dazu führen, dass Lisa mehrere kürzere Schlafphasen über den Tag verteilt hätte.

Der zirkadiane Prozess beeinflusst nicht nur Lissas Schlaf-Wach-Rhythmus. Er ist auch an **anderen wichtigen physiologischen Prozessen** beteiligt, wie der Ausschüttung bestimmter Hormone (z. B. Cortisol, Melatonin, Wachstumshormon). Mit dem zirkadianischen Rhythmus reguliert sich auch Lissas Körpertemperatur und ihre kognitive Leistungsfähigkeit über den Tag, also zum Beispiel, wann es ihr am leichtesten fällt, für Klausuren zu büffeln.

sich also im Laufe der Nacht. Es ist nicht unwahrscheinlich, während der Nacht aufzuwachen, denn zwischen den einzelnen Schlafphasen gibt es kurze Wachphasen. Diese Wachphasen finden sich oft nach dem Durchlaufen eines Zyklus. In der Regel sind diese Wachphasen kurz und es folgt eine Schlafphase. Dadurch gibt es am nächsten Tag häufig keine Erinnerungen mehr an das Wachsein, auch wenn eine Person explizit danach gefragt wird. Der Ablauf des Schlafes ist bei den meisten Menschen ähnlich. Unterschiede im Schlafrhythmus von zwei Personen können auch durch die Schlafregulierung erklärt werden.

Wie wird gesunder Schlaf reguliert?

Das dominierende Leitmodell für Schlafregulierung ist heute das **Zwei-Prozess-Modell**. Das Modell enthält den **homöostatischen Prozess - Faktor S** und den **zirkadianen Prozess - Faktor C**. Die Schlafintensität wird durch den homöostatischen Prozess beeinflusst, die Schlafdauer durch den zirkadianen Prozess.

Schlafintensität (homöostatischer Prozess - Faktor S) **Schlafdauer** (zirkadianer Prozess - Faktor C)

Die beiden Prozesse arbeiten manchmal zusammen und manchmal gegeneinander. Der Mensch kann durch sein Verhalten die Prozesse beeinflussen. Hat eine Person beispielsweise Stress oder Angst oder ist auf einem Langstreckenkflug, hat das eine Auswirkung auf die beiden Prozesse. Somit beeinflussen auch externe Faktoren den Schlaf.

1. Homöostatischer Prozess

Homöostase beschreibt ein **dynamisches Gleichgewicht**. Ein Beispiel für ein solches Gleichgewicht ist die menschliche Körpertemperatur, sie wird in einem homöostatischen Prozess geregelt – Abweichungen werden sofort gegen korrigiert.

Figure 2

Illustration of MANOVA interaction regarding bridging and elaborative comprehension scores.

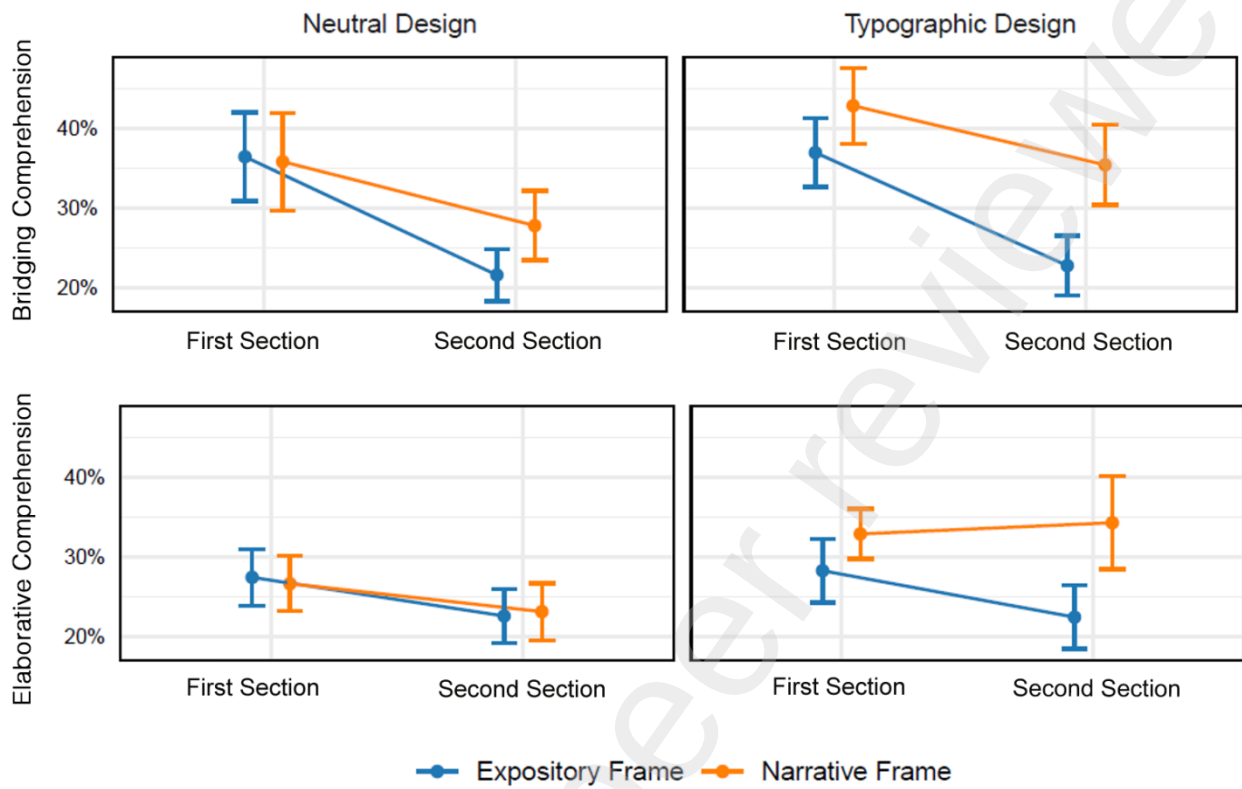


Figure 3*Illustration of sequential mediation analysis.*